

Practical Assignment
Applied AI With open source
Development Tools
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Q1.Text Tokenization using SpaCy

```
import spacy as sp

# Text Tokenization Using Spacy

text = input("Enter Text : - ")
nlp = sp.load("en_core_web_sm")
doc = nlp(text)

token = [token.text for token in doc]
print(token)
```

Output : -

Enter Text : - Hello My name is Prince Mistry
['Hello', 'My', 'name', 'is', 'Prince', 'Mistry']

Q2. Text Tokenization using NLTK.

```
from nltk.tokenize import word_tokenize ,  
sent_tokenize
```

```
# Text Tokenization Using nltk
```

```
text = input("Enter Text : - ")  
s = sent_tokenize(text)  
print(s)
```

```
w = word_tokenize(text)  
print(w)
```

Output : -

```
Enter Text : - Hello My name is Prince Mistry  
['Hello', 'My', 'name', 'is', 'Prince', 'Mistry']
```

Q3. Sentiment Analysis using TextBlob.

```
from textblob import TextBlob
```

```
text = input("Enter Text : - ")  
blob = TextBlob(text)
```

```
p = blob.sentiment.polarity  
s = blob.sentences.subjectivity
```

```
if p > 0:
    print("Positive")
elif p < 0:
    print("Negative")
else:
    print("Neutral")
```

Output : -

Enter Text : - i hate this work
Negative

Q4.Name Entity Recognition using SpaCy.

```
import spacy as sp
# Name , Entity Recognition Using Spacy
text = input("Enter Text : - ")
nlp = sp.load("en_core_web_sm")
doc = nlp(text)
for ent in doc.ents:
    print(ent.text,ent.label_)
```

Output : -

Enter Text : - india won the t20 worldcup on 29 july 2024
india GPE
29 july 2024 DATE

Q5.Simple Python-Based UI using streamlit.

```
import streamlit as st
```

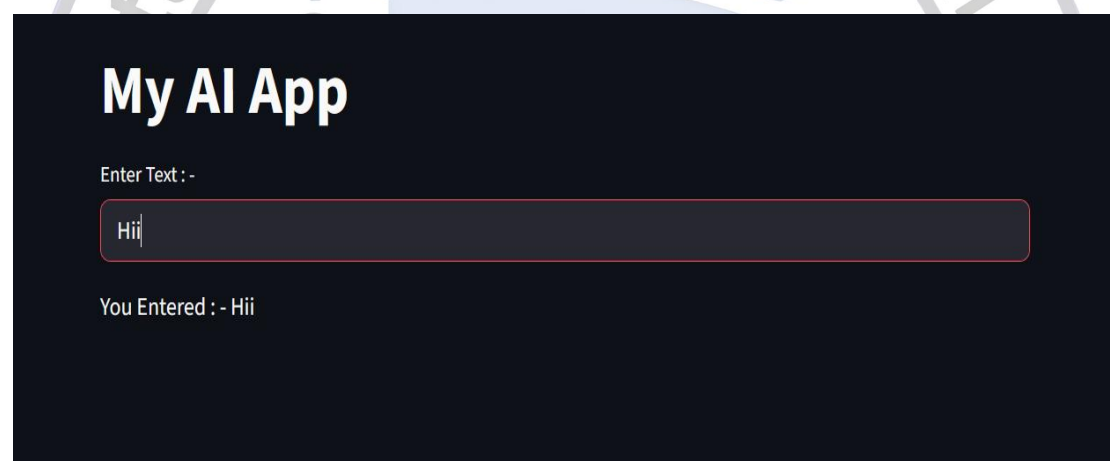
```
st.title("My AI App")
```

```
text = st.text_input("Enter Text : - ")
```

```
if text:
```

```
    st.write("You Entered" + " : - " + text)
```

Output :-



Q6.Example of Student marks predictions using streamlit & joblib.

```
import streamlit as st
```

```
from sklearn.linear_model import LogisticRegression
```

```
import joblib as jb
```

```
X =
```

```
[[19],[25],[30],[32],[33],[33],[45],[58],[68],[77],[84],[96]]
```

```
y = [0,0,0,0,1,1,1,1,1,1,1,1]
```

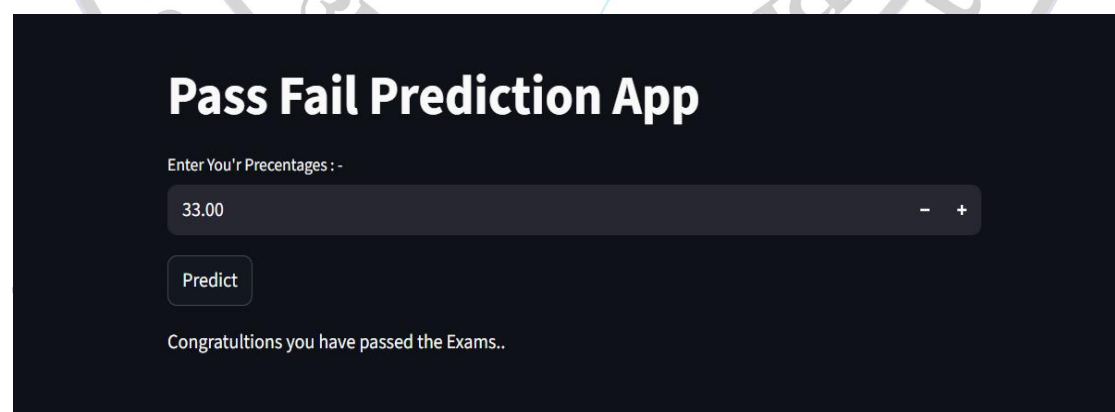
```
model = LogisticRegression()  
model.fit(X,y)
```

```
jb.dump(model,"Model.pkl")  
jb.load("model.pkl")  
st.title("Pass Fail Prediction App")
```

```
new_user = st.number_input("Enter You'r Precentages :  
- ")
```

```
if st.button("Predict"):  
    prediction = model.predict([[new_user]])  
    if prediction == 1:  
        st.write("Congratultions you have passed the  
Exams..")  
    else:  
        st.write("Better luck Next time....")
```

Output :-



Pass Fail Prediction App

Enter You'r Precentages :-

33.00 - +

Predict

Congratultions you have passed the Exams..

Pass Fail Prediction App

Enter You'r Percentages :-

Predict

Q7.Example of train a simple ML model with using streamlit & joblib.

```
import streamlit as st
from sklearn.linear_model import LinearRegression
import joblib as jb
```

```
X = [[1],[2],[3],[4],[5]]
y = [2,4,6,8,10]
```

```
model = LinearRegression()
model.fit(X,y)
```

```
st.title("Table Prediction App")
```

```
new_user = st.number_input("Enter A Number : - ")
```

```
if st.button("Predict"):
    prediction = model.predict([[new_user]])
    st.write("Predicted Number : - ", prediction)
```

Output :-

Table Prediction App

Enter A Number :-

8.00

Predict

Predicted Number :-

value

16

Q8.Example of Gradio.

import gradio as gr

```
def greed(prince):
```

```
    return "Hello , " + " " + prince
```

```
demo = gr.Interface(fn=greed,
```

```
                    inputs = gr.Textbox("Enter You'r Name : -"),
```

```
                    outputs = gr.Text("Responses : -"))
```

```
demo.launch(share=True)
```

Output :-

prince

Prince Mistry

Clear

Submit

output

Hello Prince Mistry

Flag

Q9.Simple Rule-Based ChatBot using Streamlit.

```
import streamlit as st
```

```
def chatbot(user_input):
```

```
    user_input = user_input.lower()
```

```
    rules = {
```

```
        "Hi" : "Hello , How are you !",
```

```
        "Hello" : "Hey There , How can i help you ?",
```

```
        "How are you ?" : "I am fine , hope you are fine too...",
```

```
        "What is You'r name ?" : "My name is Jarvis , Nice to meet  
you.",
```

```
        "Help" : "Tell me you'r Problem then i can help you..",
```

```
        "Bye" : "Bye have a nice day..."
```

```
    }
```

```
    for key in rules:
```

```
        if key in user_input:
```

```
            return rules[key]
```

```
    return "Sorr , I can't understand what are you saying..."
```

```
st.title("Rule-Based ChatBot")
```

```
if "chat_history" not in st.session_state:
```

```
    st.session_state.chat_history = []
```

```
user_input = input("You :- ")
```

```
if user_input:
```

```
    bot_reply = chatbot(user_input)
```

```
    st.session_state.chat_history.append(("You : -  
",user_input))
```

```
    st.session_state.chat_history.append(("Chatbot : -  
",bot_reply))
```



```
for sender,message in st.session_state.chat_history:  
    st.write(f"**{sender}** {message}")
```

Q10.Text Cleaning.

```
import string  
import re  
  
text = str(input("Enter Text :"))  
  
text_lower = text.lower()  
print(text_lower)  
  
text_punch = re.sub(r'^\w\s', "", text_lower)  
print(text_punch)  
  
text_clean = re.sub(r'\s+', ' ', text_punch).strip()  
print(text_clean)
```

Q11.Text Tokenization using NLTK.

```
from nltk.tokenize import word_tokenize,sent_tokenize  
text = "NLTK is powerful. It is widely used in NLP."
```

```
s = sent_tokenize(text)  
print(s)
```

```
w = word_tokenize(text)  
print(w)
```

Output : -

```
['NLTK is powerful.', 'It is widely used in NLP.']['NLTK', 'is',  
'powerful', '.', 'It', 'is', 'widely', 'used', 'in', 'NLP', '.']
```

